High Alert

Behavioral Stage Discovery in Addiction Treatment Reviews Using Clustering and LLM Reasoning

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NSF NRT AI Challenge

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https://github.com/ben-blake/high-alert

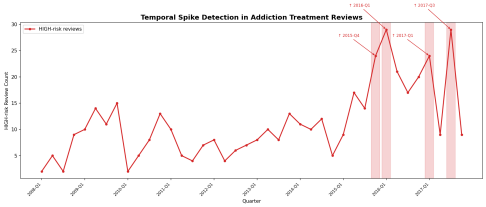
The Challenge

Addiction is a public health crisis

- 21M Americans have a substance use disorder
- Fewer than 10% receive treatment
- Relapse rates exceed 60%

The gap: clinical data misses patients who relapse and never return. But they *do* write reviews.

Goal: automatically discover recovery trajectory stages and flag risk signals from patient-generated text — *no labeled data required*.



Quarterly high-risk language spikes, 2008–2017

Dataset & Preprocessing

UCI Drug Reviews (KUC Hackathon 2018)

- 215,063 raw patient reviews
- Filtered to **7,192 reviews** matching addiction conditions and drugs
- Date range: 2008–2017 (40 quarters)
- Public data, no PII

Condition	N
Smoking Cessation	2,440
Opiate Dependence	1,477
Alcohol Dependence	518
Opiate Withdrawal	434
Alcohol Withdrawal	285

Preprocessing pipeline

1. **Clean** — HTML decode, tag strip, lowercase
2. **Embed** — `all-MiniLM-L6-v2` → 384-dim sentence vectors
3. **Keyword baseline** — 37 patterns flag HIGH / MODERATE / LOW risk
4. **Temporal** — parse dates to year-quarter for time-series analysis

Technical Methods

Three-layer approach: Keyword → Clustering → LLM Reasoning

Layer 1
Keyword Baseline

37 hand-crafted HIGH / LOW risk patterns

Fast, interpretable, no ML required

⇒

Layer 2
Semantic Clustering

UMAP (5D) + HDBSCAN → 10 behavioral clusters

Discovers populations with no labels

⇒

Layer 3
LLM Reasoning

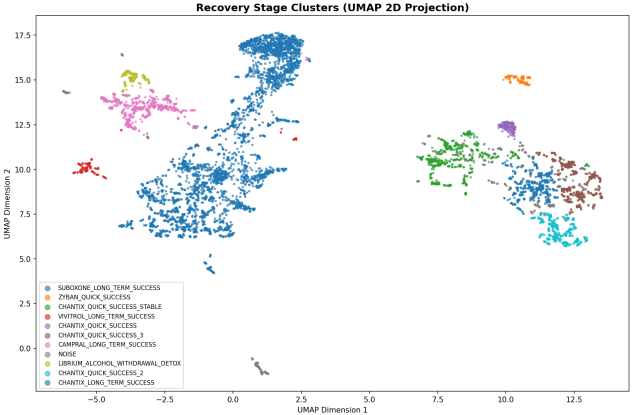
`llama-3.1-8b` via Groq

TTM stage labeling + plain-language summaries

Zero-shot, no training data

Transtheoretical Model (TTM) stages: Pre-Contemplation → Contemplation → Preparation → Action → Maintenance ↔ Relapse

Results: Recovery Stage Discovery

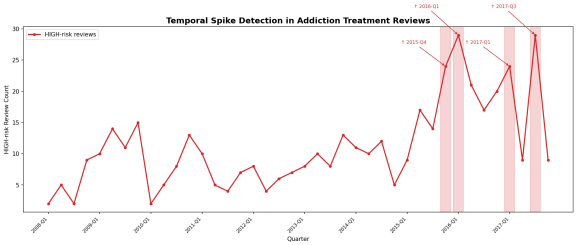


10 clusters discovered
(min_cluster_size = 130, 389 noise points)

Cluster	TTM Stage
Suboxone stable	Maintenance
Chantix quitters	Action
Vivitrol long-term	Maintenance
Librium detox	Action
Campral success	Maintenance
...	...

Key finding: survivorship bias
High-risk reviews (n=446) don't cluster — they appear as noise. Each relapse story is unique; each success story follows the same arc.

Results: Temporal Spike Detection



4 spike quarters detected
(z-score > 2.0)

- **2015-Q4** — relapse & craving spikes across opioid/alcohol drugs
- **2016-Q1** — benzo-opioid co-prescription risk awareness
- **2017-Q1** — opioid crisis peak; tapering difficulty
- **2017-Q3** — suboxone/vivitrol withdrawal distress

LLM generates a 3-sentence public health narrative per spike quarter, grounded in the most useful HIGH-risk reviews.

Evaluation

Macro-F1 vs. LLM reference labels (n = 300 reviews)

Approach	HIGH	LOW	Macro
Keyword Baseline	0.021	0.036	**0.288**
Cluster-Based	0.000	0.027	0.043

Takeaway: The two approaches are *complementary*:

- Keywords find isolated high-risk signals
- Clustering reveals dominant behavioral populations
- Neither alone is sufficient

Risk distribution (baseline classifier)

Level	N	%
MODERATE	4,908	68.2%
LOW	1,838	25.6%
HIGH	446	6.2%

The 68% MODERATE majority reflects the genuine uncertainty in addiction recovery language — neither clearly succeeding nor failing.

Live Demo

Streamlit Dashboard

Four interactive tabs:

Overview
Corpus stats, risk bar chart, approach comparison table

Recovery Stages
Interactive UMAP, LLM cluster summaries

Temporal Analysis
Spike chart, expandable LLM narratives, stage drift

Drug Comparison
Multi-select drug trends, median rating over time

```
streamlit run app.py
```

Conclusion & Future Work

What we built

- End-to-end pipeline: raw CSV → analyst briefings
- 10 behavioral clusters, TTM-mapped, LLM-summarized
- Temporal spike detection with LLM narratives
- Fully reproducible: `bash reproduce.sh`

Key finding

Survivorship bias means clustering and keywords are *complementary*, not competing approaches.

Future work

- Integrate Reddit / social media where high-risk voices are louder
- Use larger LLM (70B) for richer cluster characterization
- Train lightweight supervised classifier on LLM-labeled reviews
- Extend to real-time monitoring of emerging drug trends

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